

Pratik JOSHI

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EDUCATION

- May 2023 | **CARNEGIE MELLON UNIVERSITY - SCHOOL OF COMPUTER SCIENCE, Pittsburgh, PA, USA**
Master of Science in Intelligent Information Systems, Language Technologies Institute | CGPA : 4.12/4.0
Relevant courses : Multimodal ML, Advanced NLP, Convex Optimization, Multilingual NLP, On-Device ML
- Jun. 2019 | **BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE PILANI, Goa, India**
Bachelor of Engineering (Honors) in Computer Science | CGPA : 8.44/10.00
Relevant courses : Machine Learning, Neural Networks, Artificial Intelligence, Data Structures and Algorithms

RELEVANT EXPERIENCE

- Current | **GOOGLE DEEPMIND, Mountain View, CA**
Jun. 2023 | *Senior Research Engineer, Post-Training for Code*
 - > Core member of Gemini Mid-training (previously Code Strike) team. Focusing on SFT and Offline-RL. RL captain for Flash-size modeling.
 - > Co-created the unified agentic tool-calling infrastructure currently used across GDM for code multi-step RL, SFT distillation, and agentic evals (SWE-Bench Pro, Webdev evals). Led to state-of-the-art improvements in harness robustness, agentic capabilities in Gemini 3.
 - > Core contributor to [Gemini 3.5](#), [Gemini 3.0](#), [Gemini 2.5](#), [Gemini 1.5 Pro](#), [Gemini 1.0](#), improving code reasoning and tool-use in post-training.
 - > Core team member for [Jules](#), improving model performance for launch checkpoint via SFT distillation.
 - > Contributor to [CodeGemma](#), improving evaluations for code performance benchmarking.
- Aug. 2022 | **DEEPMIND, Mountain View, CA**
May. 2022 | *Research Engineer Intern*
 - > Trained and implemented large language models for code generation ([AlphaCode](#) style models), aiming for deployment in production IDEs.
 - > Explored multi-task training and reinforcement learning objectives for sampling and re-ranking.
- Jul. 2023 | **CARNEGIE MELLON UNIVERSITY, Pittsburgh, PA**
Sep. 2021 | *Research Assistant | Advisor : [Prof. Louis Philippe Morency](#), Mentor : [Chaitanya Ahuja](#)*
 - > Devised novel technique to prevent catastrophic forgetting and enable continual learning for multimodal gesture generations with GANs. [ICCV'23]
 - > Designing meta-learning strategies for data-scarce multimodal task scenarios.
 - > Implemented custom replay-based continual learning techniques for generative models for image generation.
- Sep. 2020 | **GOOGLE RESEARCH, Bangalore, India**
Aug. 2021 | *Research Engineer | Manager : [Dr. Partha Talukdar](#)*
 - > Developed entity-aware translation architecture for multilingual subtitle generation of technical videos; received 9/10 score from manual raters and translated 100 hours of video content for an educational organization ([NPTEL](#)).
 - > Trained neural models for temporal parsing of multilingual queries and created benchmark for model evaluation.
- Aug. 2019 | **MICROSOFT RESEARCH, Bangalore, India**
Aug. 2020 | *Research Intern | Managers : [Dr. Kalika Bali](#), [Dr. Monojit Choudhury](#)*
 - > Probed natural language inference capabilities of pre-trained language models (BERT, RoBERTa). [CoNLL'20]
 - > Analyzed resource disparity for 2500 languages, and published strongly cited paper (200+ citations). [ACL'20]
- Jul. 2018 | **MICROSOFT RESEARCH, Bangalore, India**
Dec. 2018 | *Research Intern | Managers : [Dr. Navin Goyal](#), [Dr. Monojit Choudhury](#)*
 - > Implemented neural semantic parsing techniques for converting natural language queries to code (Regex, SQL), which performed competitively with state-of-the-art techniques.

SELECTED PUBLICATIONS

- Gemini Team Google : ..., **Pratik Joshi**,... “Gemini : A Family of Highly Capable Multimodal Models”, [pdf], Coverage : [Google Blog](#), [Tech Crunch](#), [Wired Magazine](#), [NY Times](#)
- **Pratik Joshi***, Sebastin Santy*, Amar Budhiraja*, Kalika Bali, Monojit Choudhury, “The State and Fate of Linguistic Diversity and Inclusion in the NLP World”, **ACL’20** [pdf],[website], Coverage : [US FTC Commissioner Remarks](#), [NLP with Friends](#), [NLP Beyond English](#), [Quartz](#), [LacunaFund](#), [SIGTYP Newsletter](#)
- Chaitanya Ahuja, **Pratik Joshi**, Ryo Ishii, Louis-Philippe Morency, “Continual Learning for Personalized Co-speech Gesture Generation”, **ICCV’23** [pdf][website]
- **Pratik Joshi***, Somak Aditya*, Aalok Sathe*, Monojit Choudhury, “TaxiNLI : Taking a Ride up the NLU Hill”, **CoNLL’20** [pdf],[website][code]
- **Pratik Joshi**, Christain Barnes, Sebastin Santy, Simran Khanuja, Sanket Shah, Anirudh Srinivasan, Satwik Bhattamishra, Sunayana Sitaram, Monojit Choudhury and Kalika Bali, “Unsung Challenges of Building and Deploying Language Technologies for Low Resource Language Communities”, **ICON’19** [pdf][slides]
- Basil Abraham, Danish Goel, Divya Siddarth, Kalika Bali, Manu Chopra, Monojit Choudhury, **Pratik Joshi**, Preethi Jyoti, Sunayana Sitaram and Vivek Seshadri, “Crowdsourcing Speech Data for Low-Resource Languages from Low-Income Workers”, **LREC’20** [pdf], Coverage : [Indian Express](#), [Microsoft Stories](#)
- Sanket Shah, **Pratik Joshi**, Sebastin Santy, Sunayana Sitaram, “CoSSAT : Code-Switched Speech Annotation Tool”, **EMNLP’19** [pdf]
- Anshul Bawa, Pranav Khadpe, **Pratik Joshi**, Kalika Bali and Monojit Choudhury, “Do Multilingual Users Prefer Chat-bots that Code-mix? Let’s Nudge and Find Out!”, **CSCW’20** [pdf]

SELECTED WORK MENTIONS AND MEDIA COVERAGE

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| 01/2024 | Tech Crunch on “Google Gemini : Everything you need to know about the new generative AI platform”, [link] |
| 12/2023 | The New York Times on “Google Updates Bard Chatbot With ‘Gemini’ A.I. [...]”, [link] |
| 12/2023 | BBC on “Google claims new Gemini AI ‘thinks more carefully’ ”, [link] |
| 12/2023 | CNN on “Google launches Gemini, its most-advanced AI model yet [...] ”, [link] |
| 05/2023 | Centre for Democracy and Technology on “Lost in Translation : Large Language Models in Non-English [...]”, [link] |
| 01/2023 | Indian Express on “How Microsoft’s Project ELLORA is helping small languages like Gondi become [...]”, [link] |
| 01/2023 | Microsoft Stories on “Microsoft Research project helps languages survive — and thrive”, [link] |
| 08/2022 | U.S. Federal Trade Commission (FTC) on “Who is being left behind? Enforcement Priorities for a Tech [...]”, [link] |
| 04/2022 | The Gradient on “New Technology, Old Problems : The Missing Voices in Natural Language Processing”, [link] |
| 04/2020 | SIGTYP on “Recent Developments in Computational Typology & Multilingual NLP”, [link] |

TALKS AND PANELS

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| 2025 | Innovation Hacks Hackathon at Arizona State University on “AI in Innovation”, <i>Remote</i> |
| 2024 | Lee Language Lab Talk Series on “The State and Fate of Linguistic Diversity and Inclusion in the NLP World”, <i>Remote</i> [Link] |
| 2023 | Centre for Democracy and Technology Panel on “The Nuts and Bolts (and Nuances) of Foundation Models”, <i>Remote</i> [Abstract] [Link] |
| 2021 | Google Research on “NLP for the Long Tail”, <i>Remote</i> |
| 2020 | NLP with Friends on “State and Fate of Linguistic Diversity in the NLP World”, <i>Remote</i> [Abstract][Link] |

HONORS AND AWARDS

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| 2025 | BITS Global 30 under 30 award for Research Leadership [link] |
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ACADEMIC SERVICE AND REVIEWING

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| Reviewer | ICLR’24, LREC-COLING’24, ACL’24, NAACL’24 |
| Subreviewer | LREC’20, ACL’20, ICON’19 |

TEACHING AND MENTORING

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| May 2018 | MACHINE LEARNING BITS PILANI, Goa, India |
| Jan’ 2018 | Teaching Assistant Course Instructor : Prof. Ashwin Srinivasan |
| | ➤ Created coding assignments and final project for Machine Learning (BITS F464). [course material] |